PPC MINI PROJECT

QUANTUM GATE SIMULATION USING C PROGRAMMING



STATEVECTOR SIMULATOR & PQC CLASSIFIER

PPC CA 3

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INTRODUCTION

PROBLEM STATEMENT & MOTIVATION

WHAT THIS PROJECT DOES

SIMULATE
Runs quantum gates step-by-step

EXPLAIN
Shows state changes after each gate

RUN INPUT
Takes manual input or file input

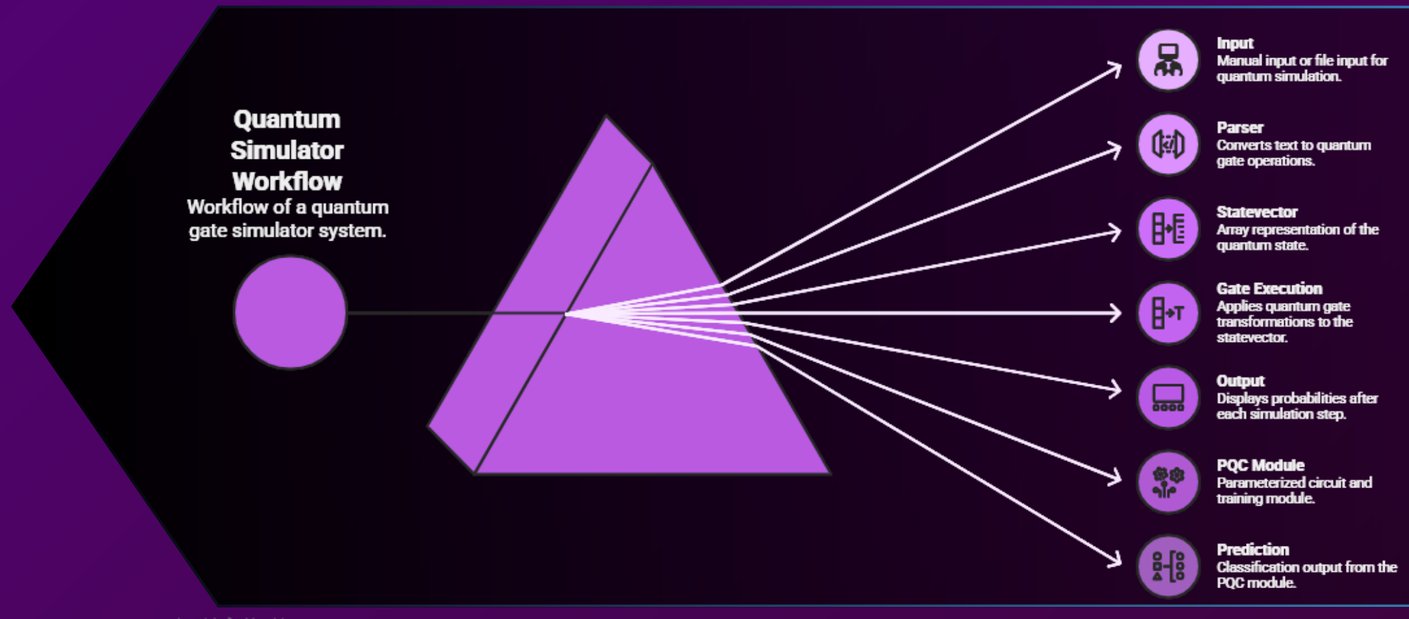
LEARN
Uses PQC for classification

PROBLEM

- Quantum circuits act like black boxes
- High-level languages hide execution details
- Difficult to understand how gates transform states

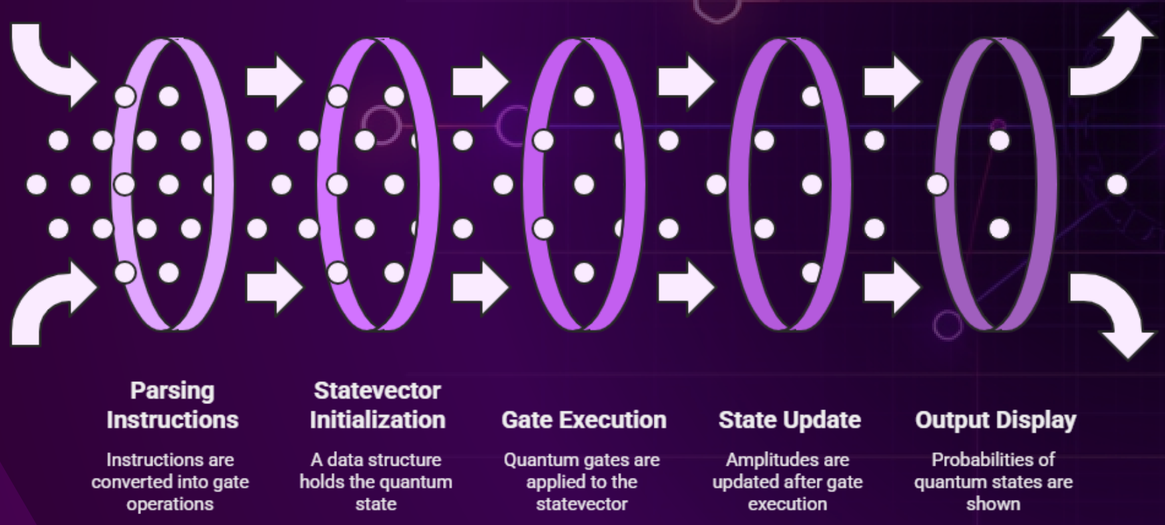
WHY THIS MATTERS:

- Hard for beginners to learn quantum concepts
- No step-by-step visibility of transformations
- Limits understanding of circuit behavior



PROGRAM ARCHITECTURE

FLOW DIAGRAM



SYSTEM COMPONENTS

- INPUT: TAKES CIRCUIT (MANUAL / FILE)
- PARSER: CONVERTS TEXT TO GATE OPERATIONS
- STATEVECTOR: STORES QUANTUM STATE AS ARRAY
- GATE ENGINE: APPLIES TRANSFORMATIONS
- (X, H, RX, RY, RZ, CX)
- OUTPUT: DISPLAYS PROBABILITIES AFTER EACH STEP

EXECUTION FLOW

- READ INPUT CIRCUIT
- PARSE GATE INSTRUCTIONS
- INITIALIZE STATEVECTOR
- APPLY GATES SEQUENTIALLY
- UPDATE STATE AFTER EACH GATE
- DISPLAY FINAL PROBABILITIES

DEMONSTRATION

GATE EXECUTION

INPUT:
RY 0 1.57
H 1
CX 0 1

STEPS:
ROTATION → SUPERPOSITION → ENTANGLEMENT

OUTPUT:
|00> ≈ 0.25
|01> ≈ 0.25
|10> ≈ 0.25
|11> ≈ 0.25

PQC CLASSIFIER

FOR XOR GATE
(NORMALLY, SHALLOW NN)

EPOCH 1 → HIGH LOSS
EPOCH 200 → LOW LOSS

PREDICTION:
[0,0] → 0
[0,1] → 1
[1,0] → 1
[1,1] → 0

KEY RESULTS

GATES APPLIED CORRECTLY
STATE UPDATES AFTER EACH STEP
EQUAL PROBABILITY DISTRIBUTION ACHIEVED

CONCLUSION

ACHIEVEMENTS

- SIMULATED QUANTUM GATES USING C
- DISPLAYED STATEVECTOR TRANSFORMATIONS
- IMPLEMENTED PQC-BASED CLASSIFICATION

ADVANTAGES

- TRANSPARENT AND EXPLAINABLE SYSTEM
- BUILT FROM SCRATCH USING CORE C CONCEPTS
- SUPPORTS BOTH INTERACTIVE AND FILE EXECUTION

LIMITATIONS

- LIMITED TO SMALL NUMBER OF QUBITS
- SIMULATION ONLY (NOT REAL QUANTUM HARDWARE)

THIS PROJECT DEMONSTRATES HOW QUANTUM CIRCUITS CAN BE UNDERSTOOD STEP-BY-STEP USING LOW-LEVEL IMPLEMENTATION.

THANK YOU ! ANY QUESTIONS?